

Discussion Notes Structure And Magnitudes

What Does the Structure of Double Entry Actually Carry?

Notes on the Modeling and Notation of "An Information Property of Double Entry Accounting"

The paper by Antle, Fellingham, Lin, and Schroeder (2025) models the double-entry accounting system as a communication channel in the sense of Shannon (1948). Accounts are two-element vectors, journal entries are 2x2 transformation matrices — specifically, the Pauli X and Z matrices from quantum information theory — and the information capacity of the system is measured by the number of distinct signals a receiver can extract. The central result is that one account can transmit two bits of information, but only when two financial statements articulate and at least two accounts satisfy both the accounting identity and the cash flow reconciliation identity.

Our discussion examined the foundations of this framework, beginning with the paper's decision to abstract away from magnitudes.

Magnitudes and the Asset/Liability Distinction

In practice, magnitudes carry enormous information. It matters greatly whether Sales was \$1,000 or \$100. The paper removes scale entirely (Assumption 1), reducing all accounts to a balance of one and focusing exclusively on the structural type of each account — asset, expense, liability, or revenue.

Is this a reasonable abstraction? We argued that it is, because the paper's question is specifically about the information-carrying capacity of the *structure* of double entry, not the data flowing through it. You cannot isolate the structural contribution without removing magnitudes. The result — two bits per account — should be understood as a floor on the system's capacity, not a ceiling.

But the discussion pushed further. The distinction between assets and liabilities is itself a magnitude phenomenon. A "negative asset" is a liability. Interest rate swaps illustrate this directly: the same contract sits on either side of the balance sheet depending on the sign of its fair value. The label is just the sign bit of a continuous variable. This observation is consistent with the paper's own Definition 2(ii), which treats accounts on the same financial statement as indistinguishable — scalar multiples of one another.

The Irreducible Structural Distinction

If asset versus liability is a magnitude distinction, what structural distinction actually survives? The accounting identity, $\text{Assets} = \text{Equities}$, separates resources from claims. These are not positive and negative versions of the same quantity. A receivable does not become equity by changing sign. This is the genuinely structural boundary in the system.

But this raised a concern: if we reduce the entire system to $A = E$ with magnitudes removed, have we arrived at a triviality? A single equation with a single structural distinction and no magnitudes seems to carry very little information.

The answer is that the information capacity does not come from the identity alone. It comes from the requirement that the same economic events be recorded on *multiple articulating statements* — the accrual statement and the cash flow statement — in structurally different but reconciled ways. One identity is trivial. Two constraints that transactions must simultaneously satisfy create the redundancy that a receiver can exploit to distinguish among different types of events.

The Decomposition of Equities

The discussion then identified what may be the deepest structural feature at work. Equities are not a monolithic category. They decompose into two fundamentally different sources: amounts contributed by owners and amounts generated by transactions and events. This is not a difference in magnitude — it is a difference in kind. A \$20 increase in equity from a capital contribution is a categorically different event from a \$20 increase arising from \$30 in revenue less \$10 in expenses, even though the net magnitude is identical.

The income statement exists precisely to make this distinction. It is the system's way of recording not just *what* the balances are but *why* they changed. The "why" is encoded in the classification structure: which accounts were debited and credited, whether the change flowed through operations or came directly from owners, whether a cash expenditure was capitalized or expensed. None of this is magnitude information. All of it is informative.

This is, we believe, the strongest articulation of what the double-entry structure contributes beyond magnitudes. The structure adds channel capacity because it decomposes changes in wealth into their sources and requires consistency across multiple representations of the same underlying activity. The paper's formal results — the two-bit capacity, the necessity of articulation, the role of both X and Z operators — are the mathematical expression of this accounting insight.